

Deep Learning Segmentation of Thermo-Diffusively Unstable Premixed Spherical Flames

Master's Thesis by

Orso Alexander Ruben Birelli Schmid

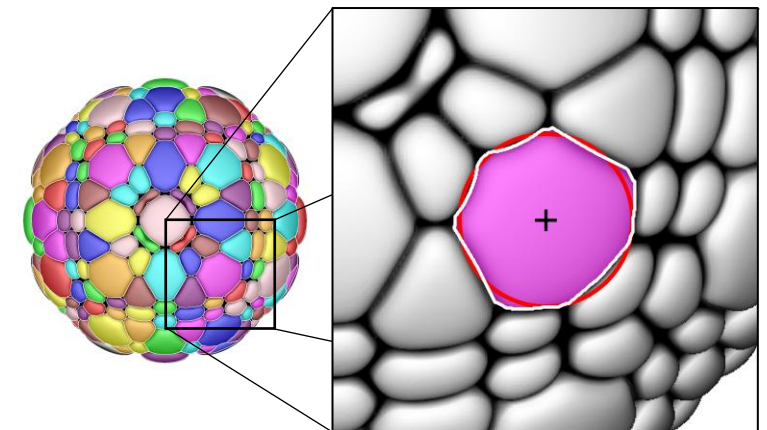
Zürich, 17th July 2025

Supervisor

Dr. Christos Frouzakis

Professor

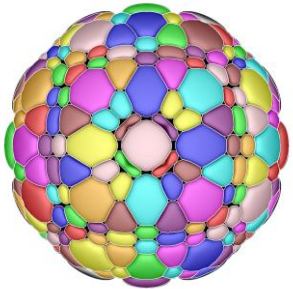
Prof. Dr. Nicolas Noiray



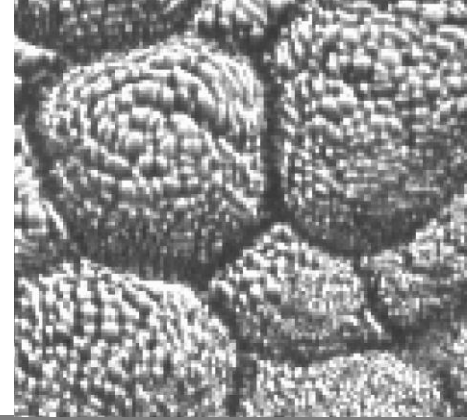
Agenda



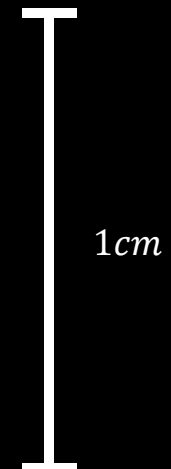
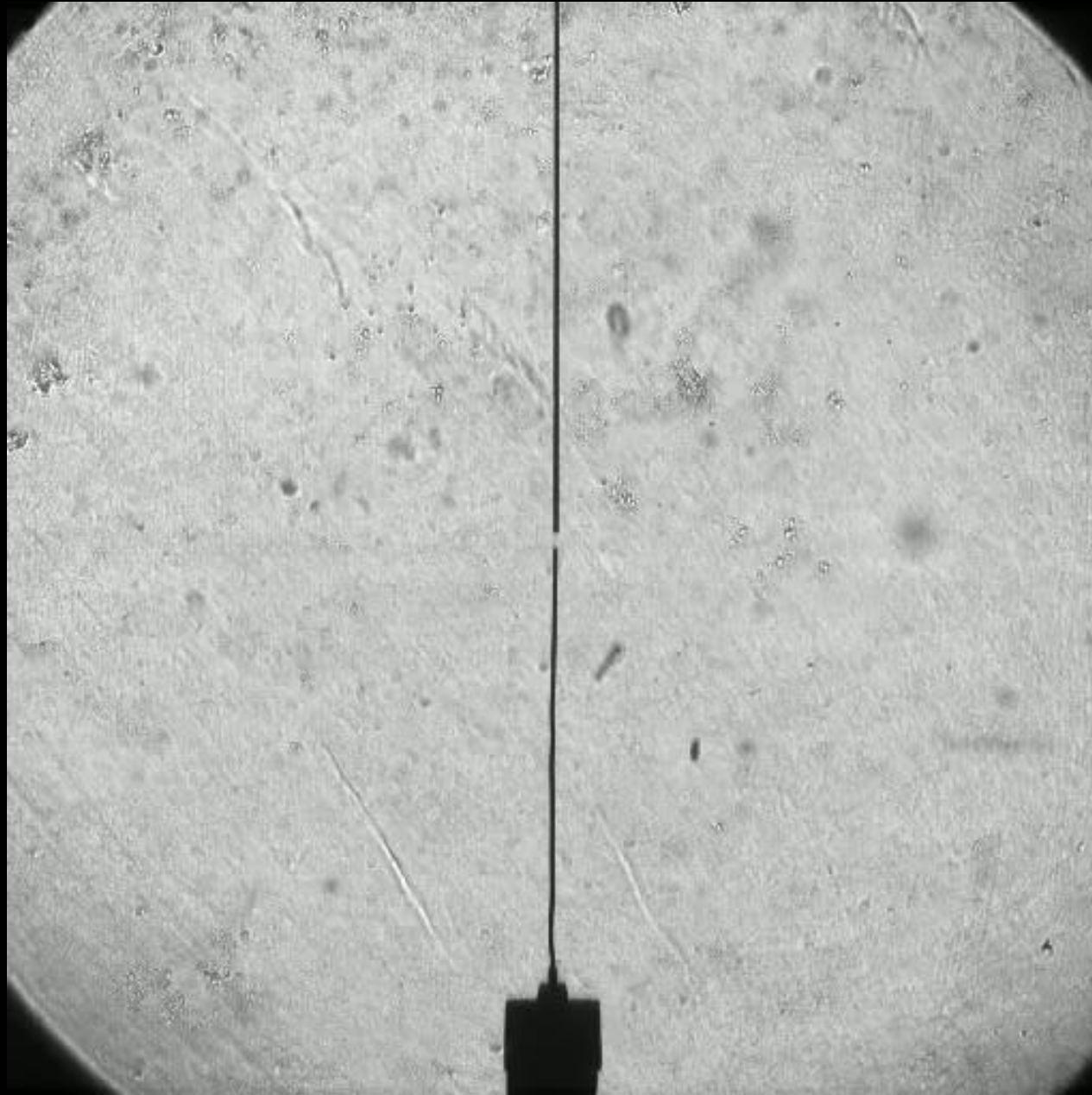
- ▶ Spherical Expanding Flame Physical Background
- ▶ Measurements



- ▶ DNS Simulation
 - ▶ Integrated Analysis Pipeline
 - ▶ Results
-
- ▶ Conclusion



Spherical Expanding Flame Physical Background



$\approx 3ms$ total

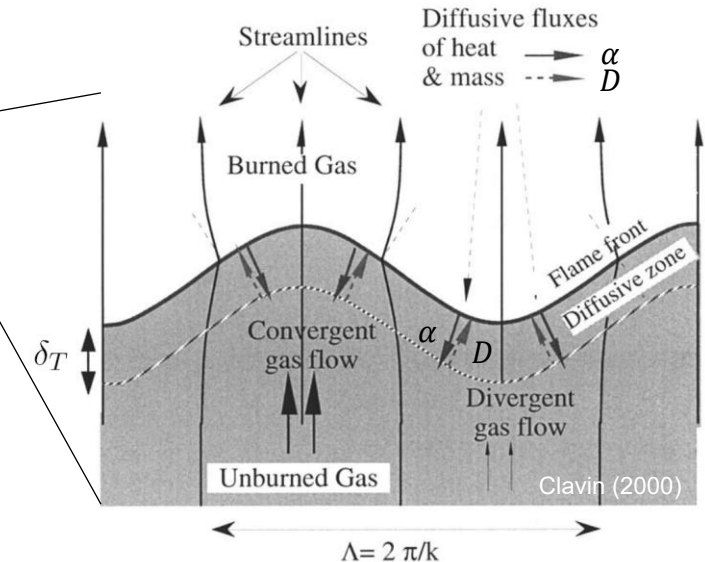
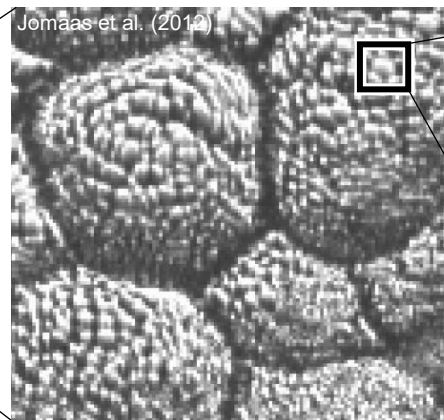
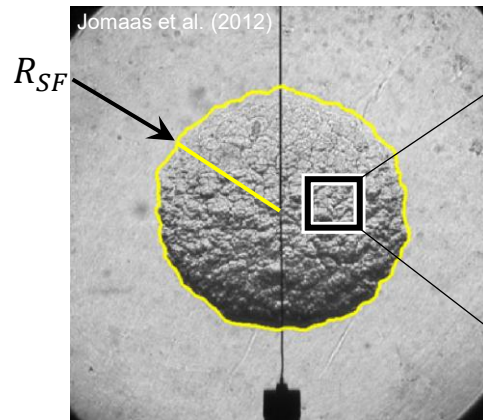
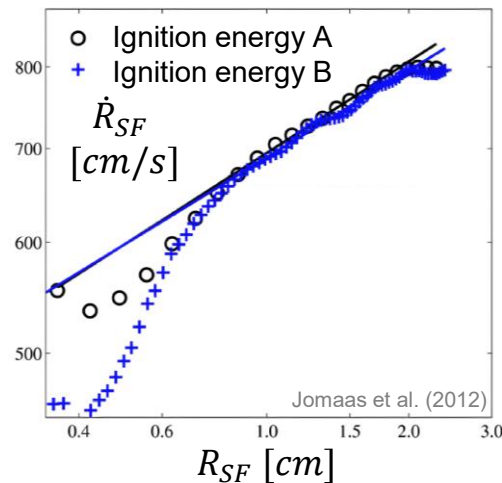
premixed lean H₂-air

$\phi = 0.6$

$p = 5atm \approx const.$

Jomaas et al. (2012)

Expanding Spherical Flames (SF)



- Self-acceleration
- Self-turbulisation

Expanding spherical flame

- Spark ignition
- Explosion hazards
- Experimentally accessible

Wrinkling, cells

- Emergent structures
- Potentially self-similar (fractal)

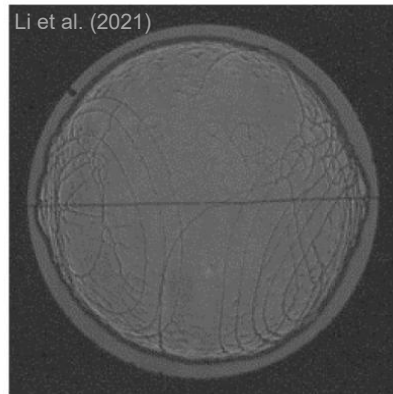
Intrinsic instabilities

- hydrodynamic $\sim \sigma = \frac{\rho_u}{\rho_b} \gg 1$
- thermo-diffusive $\sim Le = \frac{\alpha}{D} \ll 1$
- Area generation

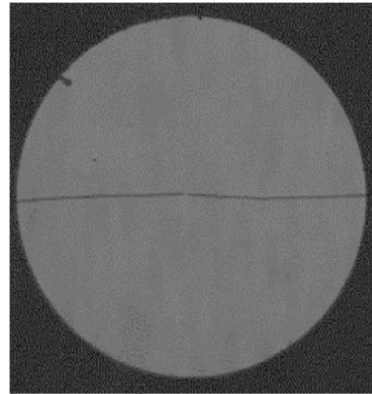
How to measure cellular structures?

Measurements in the Past and Now, using Deep Learning

Measuring Radii and Cells – Classical Image Processing

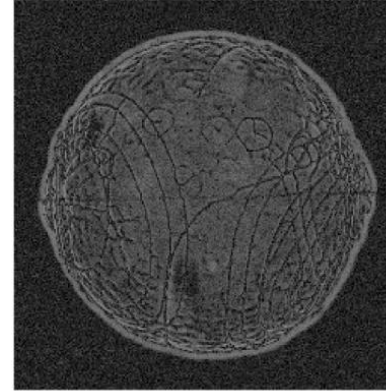


Raw flame image



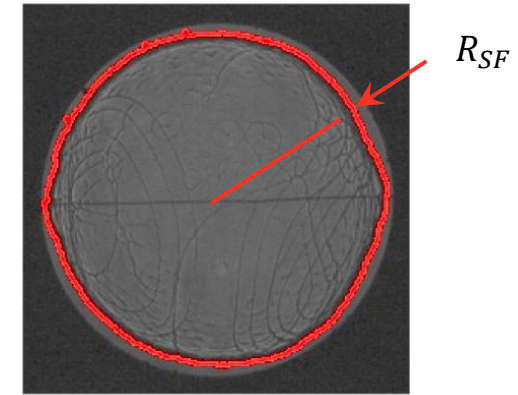
Background Image

=



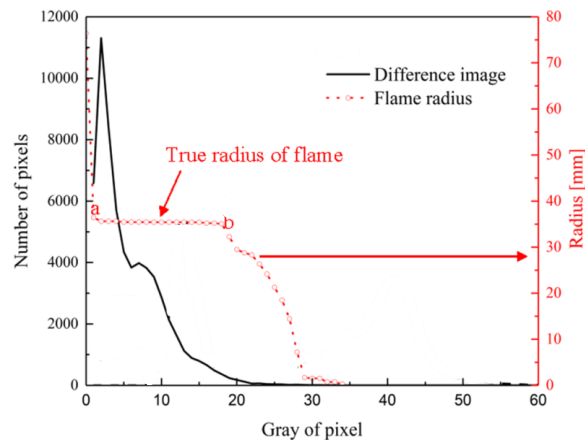
Difference image

Thresholding
Binarisation



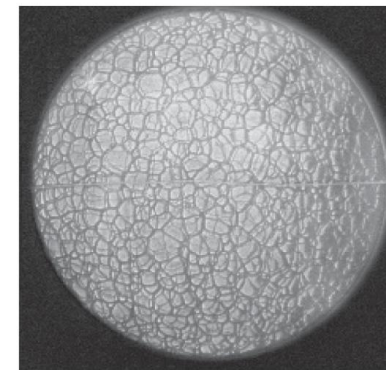
Detection result

Thresholding Binarisation



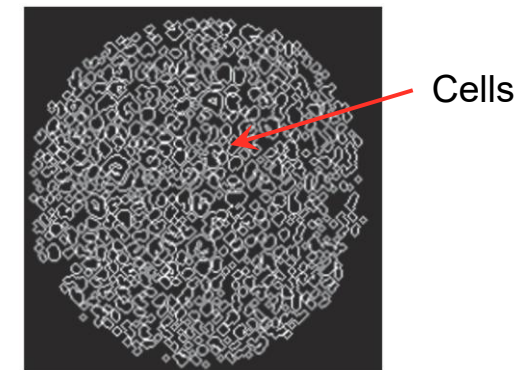
Issues:

- ▶ Results sensitive to user defined thresholds
- ▶ Manual parameter adaptation to imaging
- ▶ Sensitive to noise
- ▶ Cells often merged



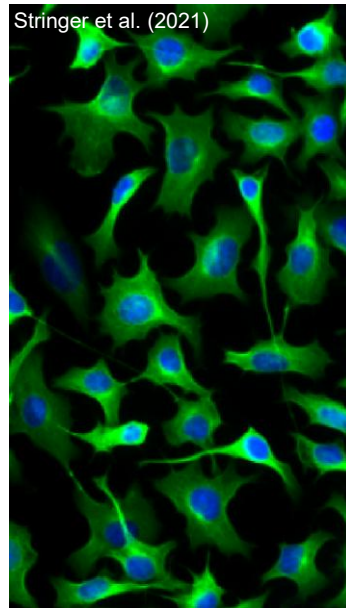
Flame image after preprocessing

Thresholding
Binarisation
+
Connected
Components

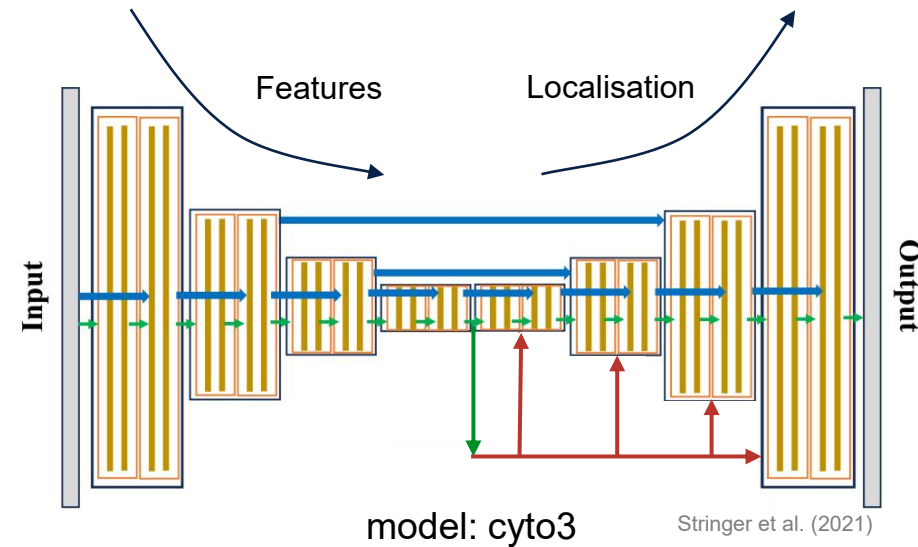


Cell-outlines

Measuring Radii and Cells – Deep Learning Segmentation



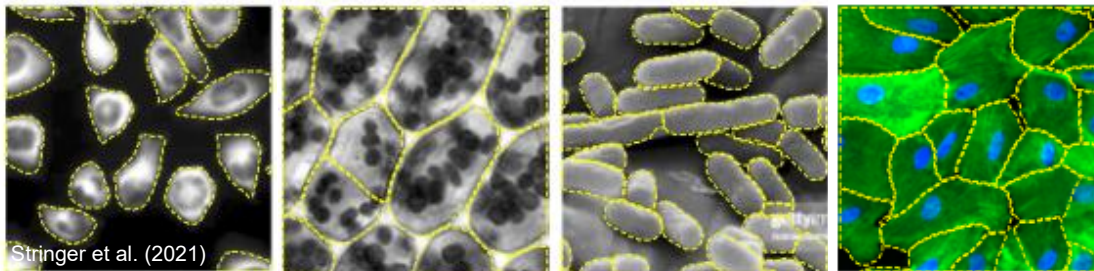
Original image



Algorithm: Cellpose U-Net

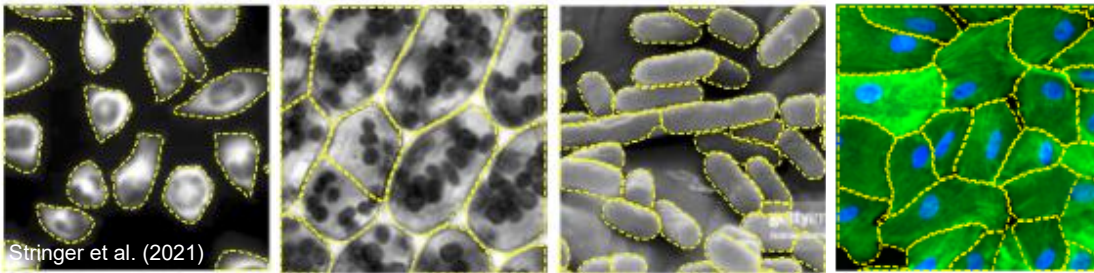
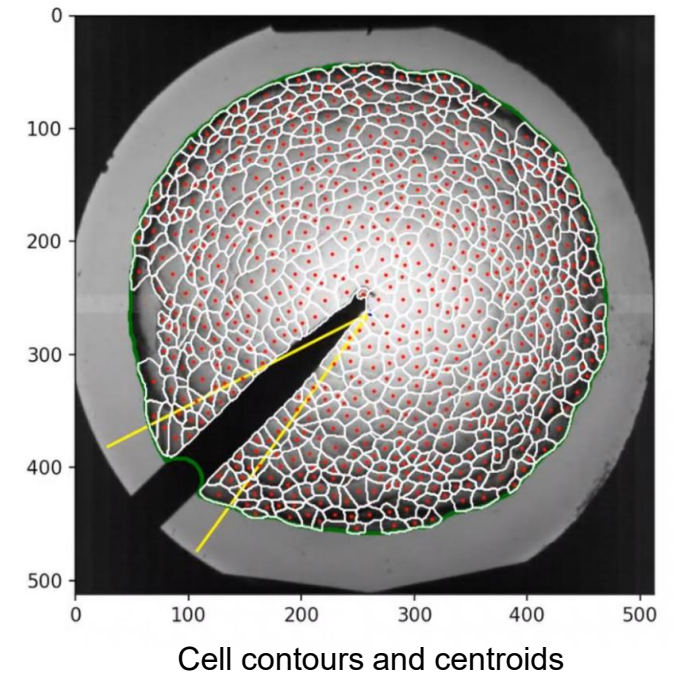
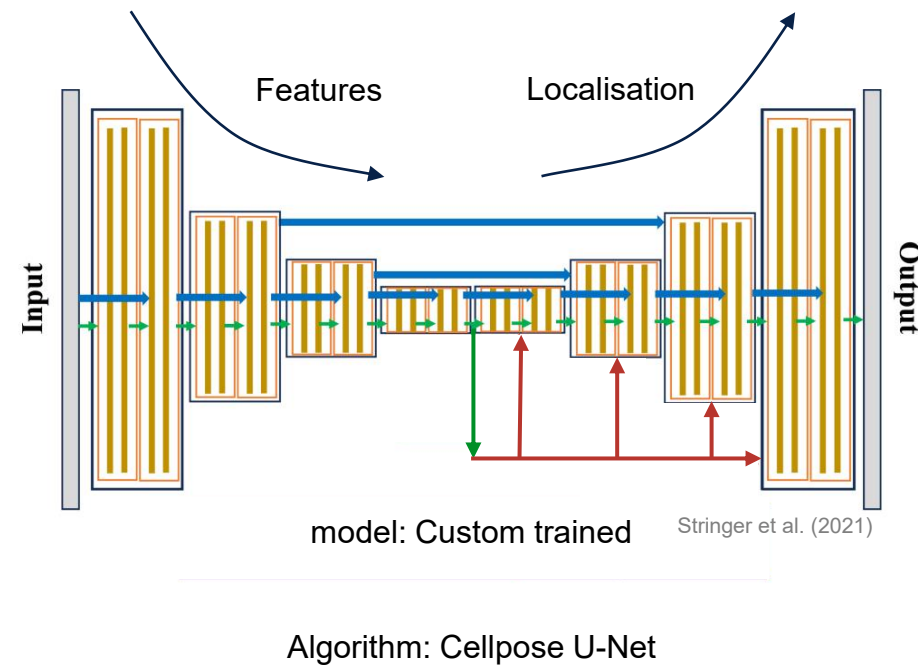
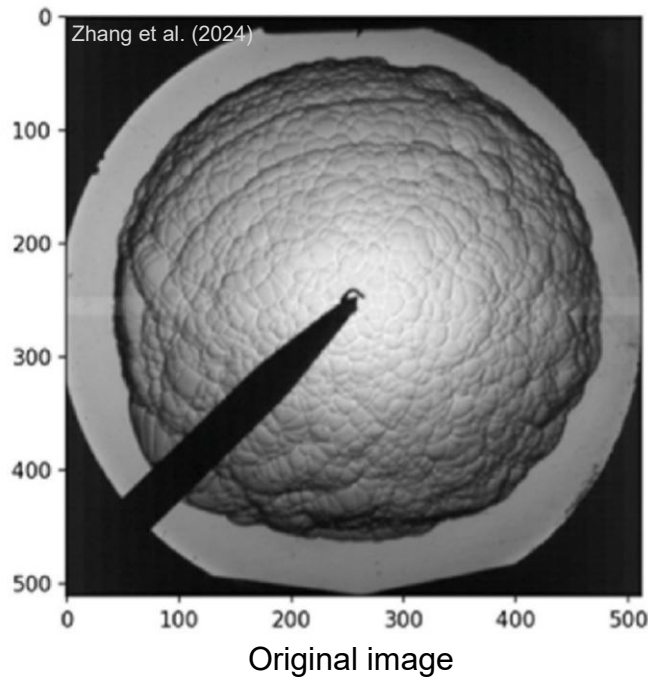


Segmented Cell



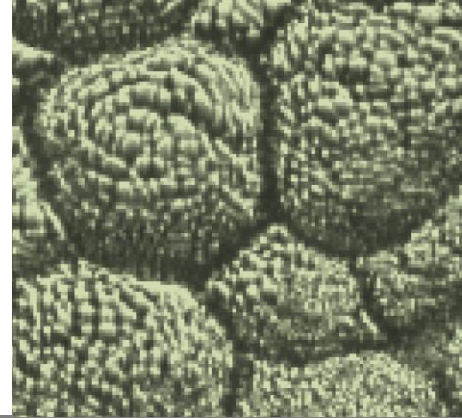
- ▶ Algorithm developed for use in biology
- ▶ Generalist models (weights) trained from manual segmentation
- ▶ *cyto3* trained on ca. 8000 images in many (bio) cell types
- ▶ Custom training possible

Measuring Radii and Cells – Deep Learning Segmentation



- ▶ Algorithm developed for use in biology
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- ▶ Custom training possible

Simulation



Direct Numerical Simulation (Altantzis 2011)

Method:

- Spectral element solver based on Nek5000 from ANL

Formulation:

- Low *Mach number* reactive flow plugin (filters acoustics) from LAV/ETHZ Kerkemeier (2010)

Numerical setup - Spherical expanding flame:

- Lean H2-air $\phi = 0.6$
- Detailed chemistry: 9 species and 19 elementary reactions by Li et al. (2004)
- Mixture average transport.
- No gravity, no radiation, no Soret effect

Mesh:

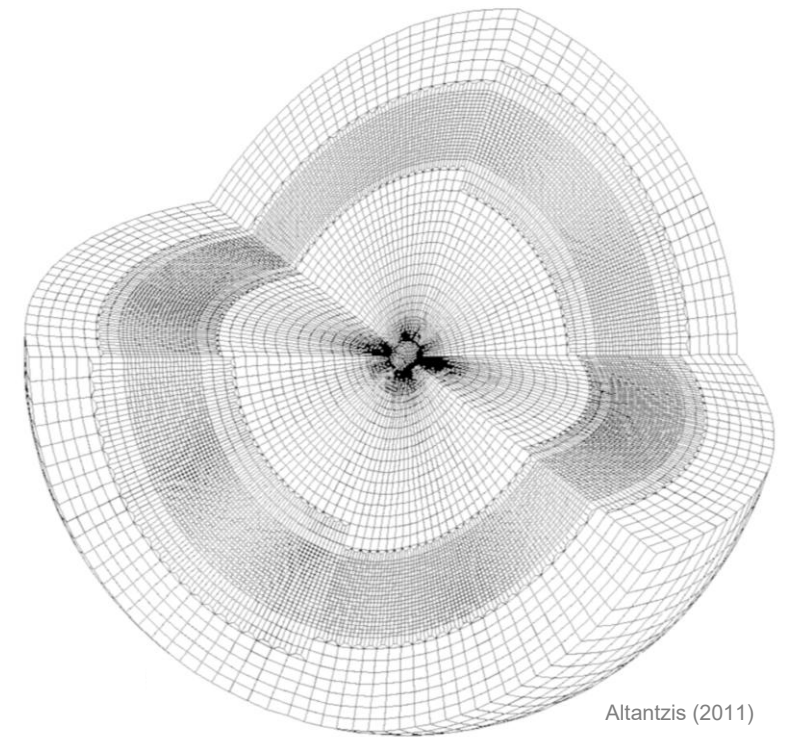
- Resolved flame structure with 11 pts/ δ_T
- 3D spherical mesh ($\approx 900'000$ elements)
- Manual refinement around the flame

Boundary condition:

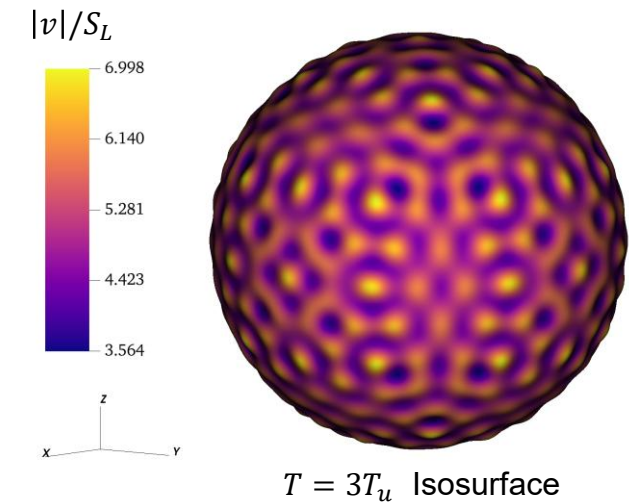
- Inner ($R_{in} = 5\delta_T$): $v = \nabla T = \nabla Y_k = 0$
- Outer ($R_{in} = 100\delta_T$): outflow ($\nabla(\cdot) = 0$)

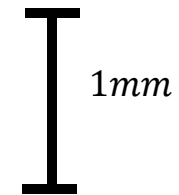
Initial condition:

- Ideal spherical premixed flame perturbed by spherical harmonics
- $T_u = 298\text{K}$, $p = 5\text{atm}$



Altantzis (2011)





$\approx 1ms$ total

premixed lean H₂-air

$\phi = 0.6$

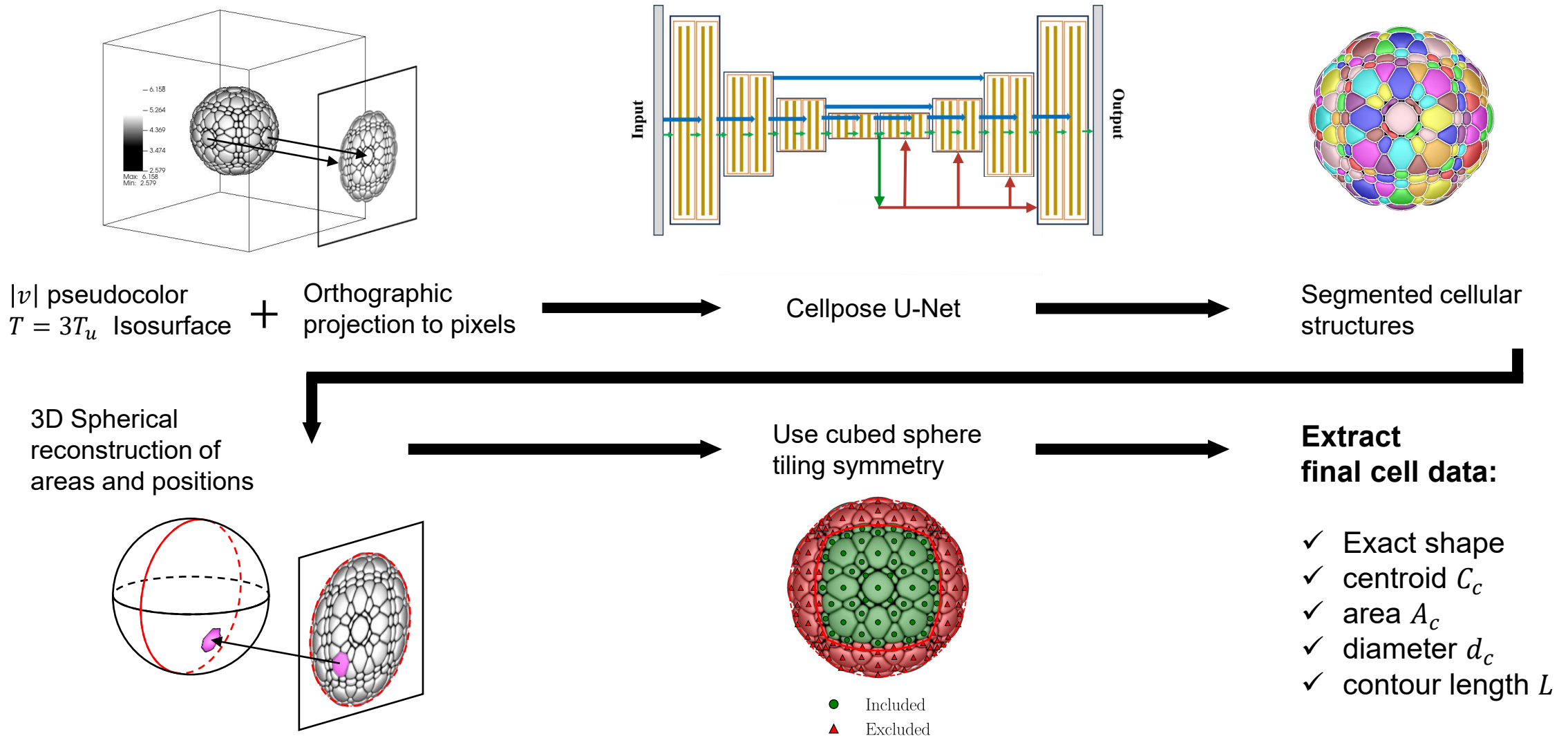
$p = 5atm \approx const.$

$T = 3T_u$ Isosurface

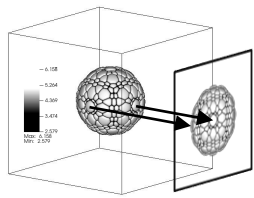
$|v|$ Pseudo-coloured

Integrated Analysis Pipeline

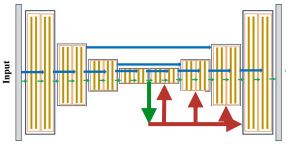
Integrated Analysis Pipeline



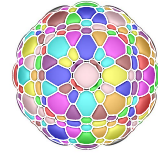
Integrated Analysis Pipeline



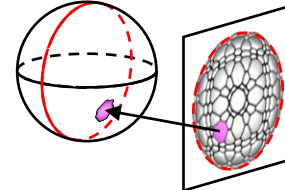
Flame surface
projection



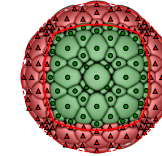
Cellpose U-Net



Segmented
cellular structures



3D Spherical
reconstruction



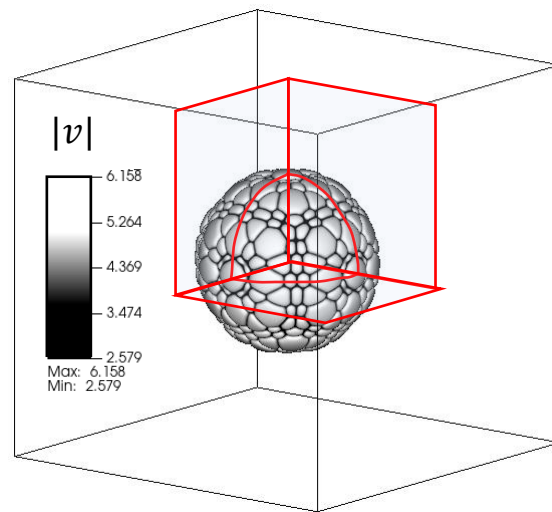
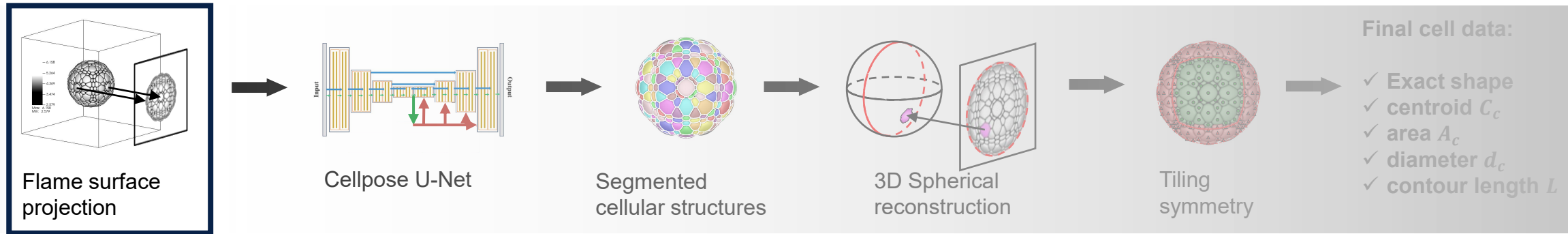
Tiling
symmetry



Final cell data:

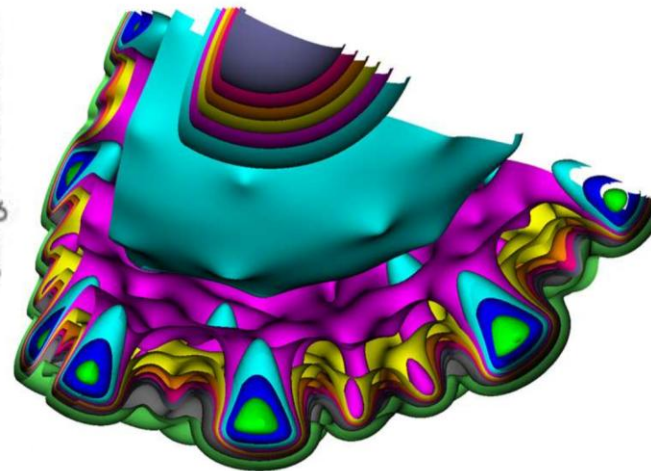
- ✓ **Exact shape**
- ✓ **centroid C_c**
- ✓ **area A_c**
- ✓ **diameter d_c**
- ✓ **contour length L**

Integrated Analysis Pipeline | Defining the Flame Surface

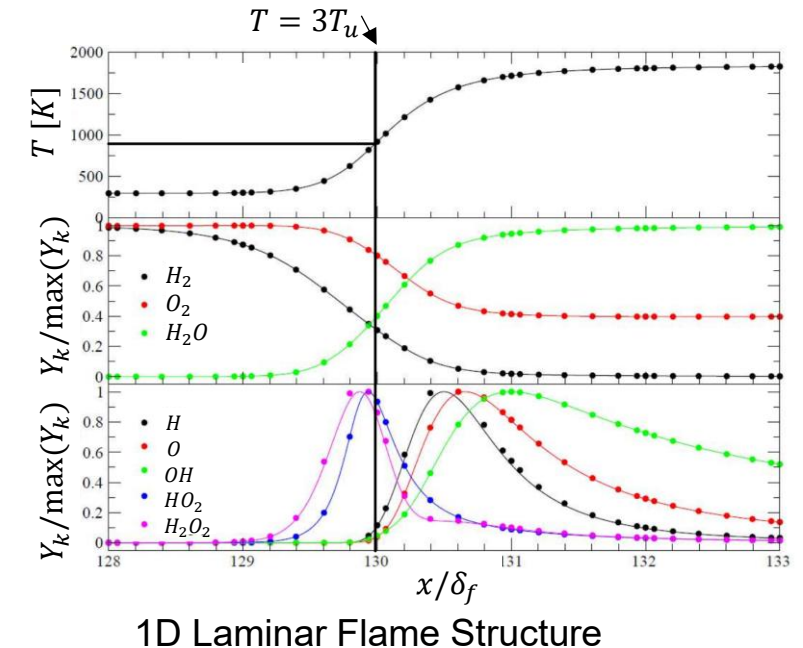


$T = 3T_u$ Isosurface

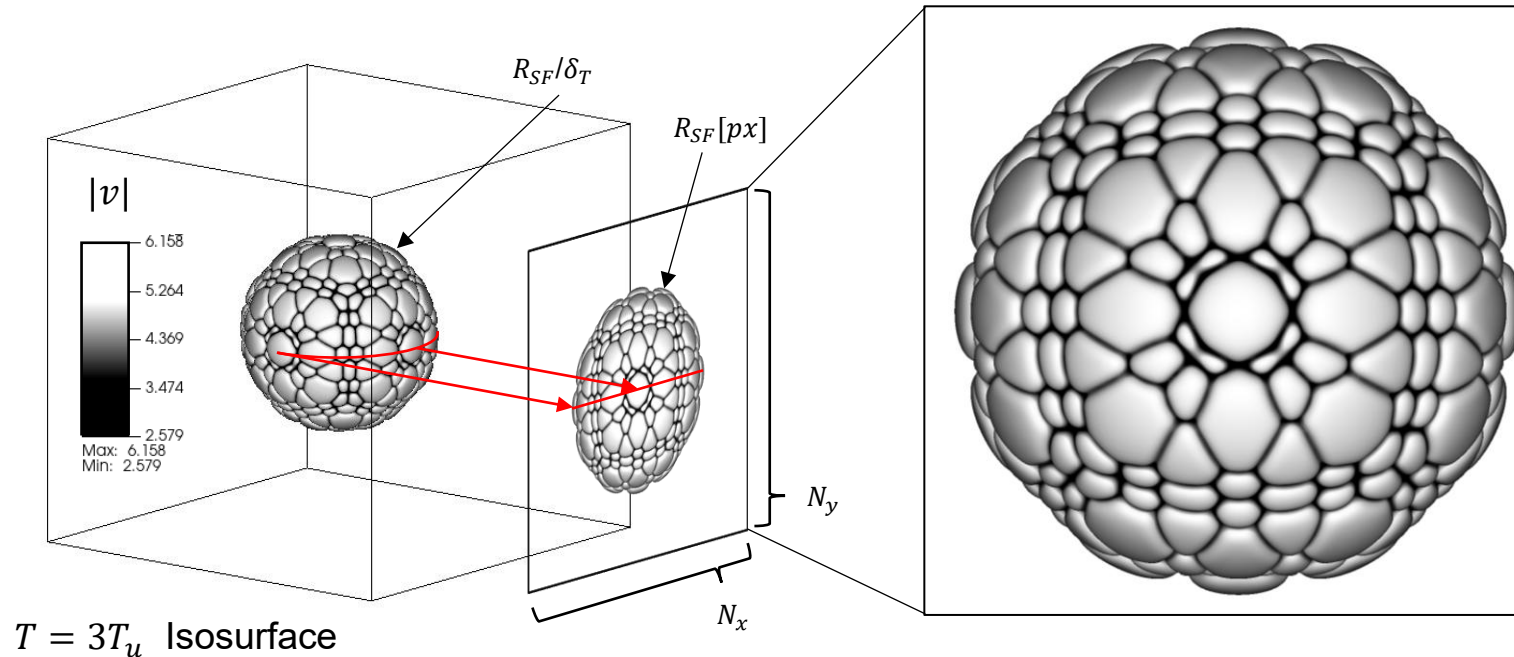
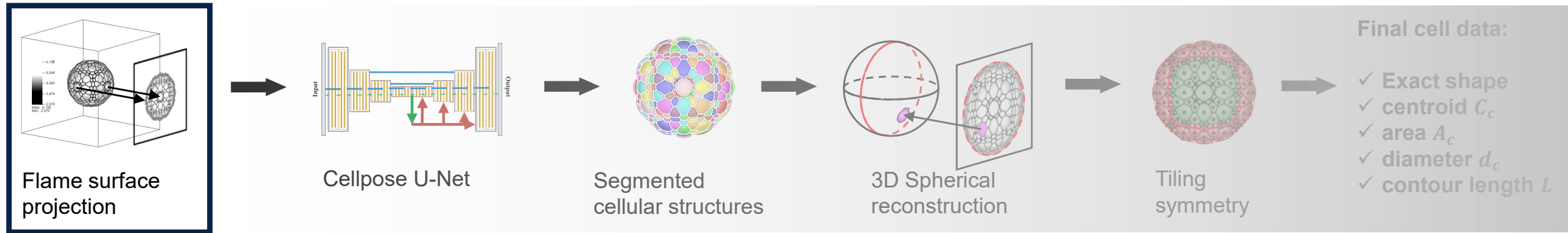
Coloring: $|v|$ indicative of the influence of curvature on the quiescent flow field
 → Try other things! OH, curvature, heat release rate



T Isosurfaces

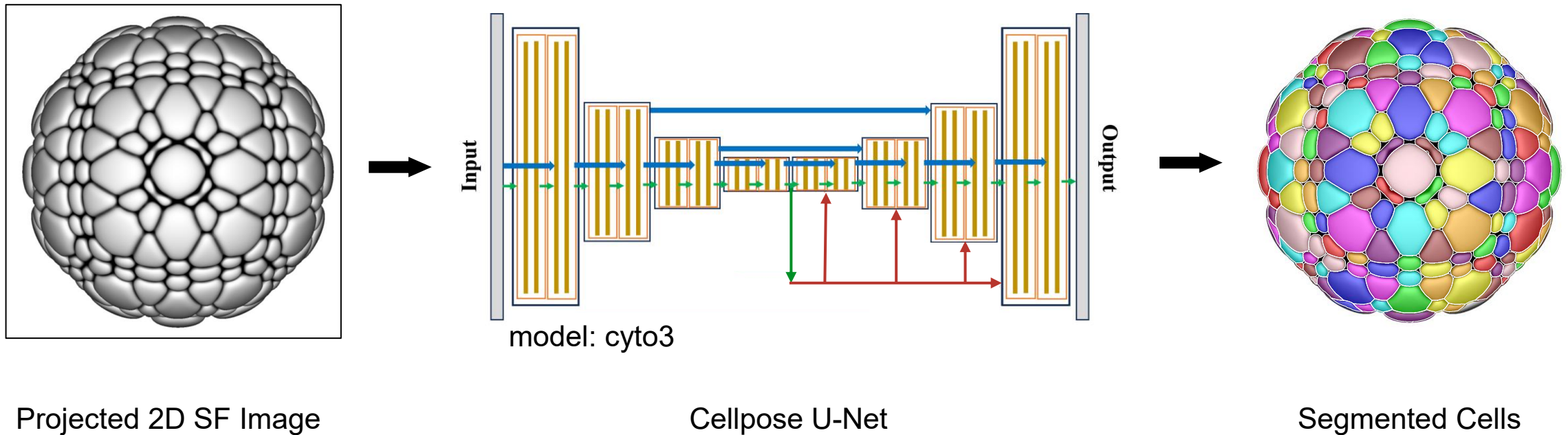
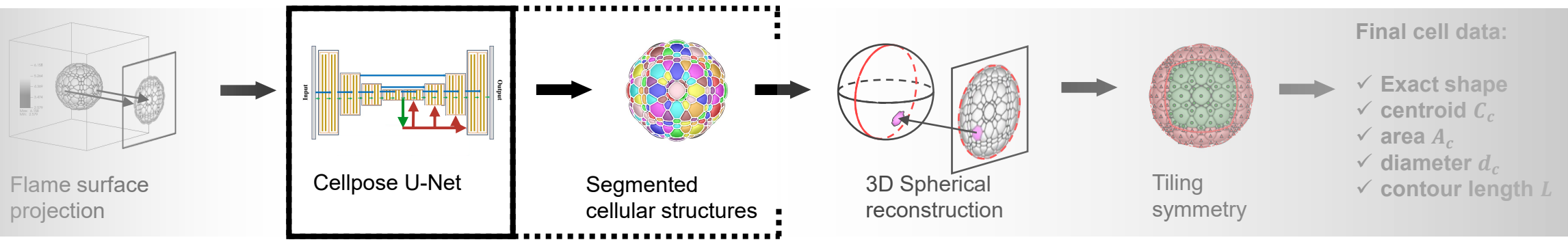


Integrated Analysis Pipeline | Generating Images to Be Segmented

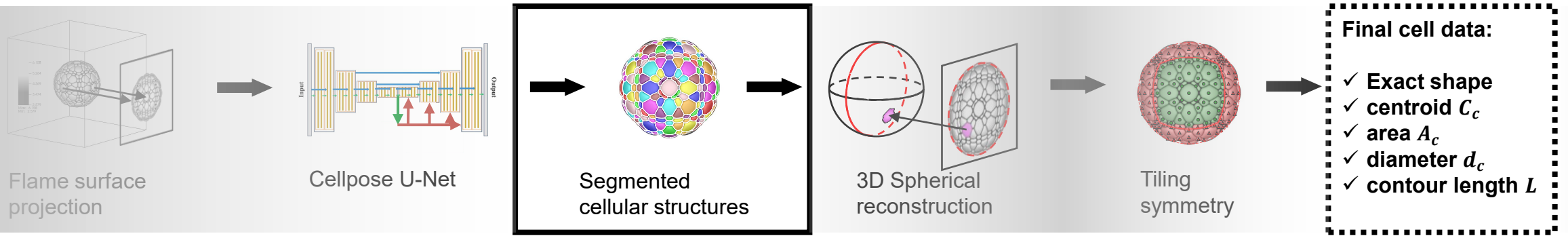


Orthographic projection to pixels

Integrated Analysis Pipeline | Cellpose U-Net & cyto3 Pretrained Model



Integrated Analysis Pipeline | Extracted Properties



Centroid:

$$C_c = \frac{1}{A_c} \sum_{Cell_c} x_i$$

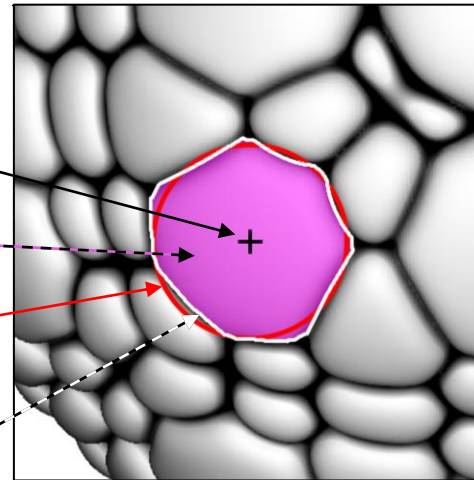
Area:

$$A_c = |Cell_c|$$

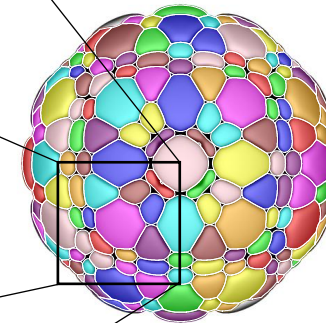
Diameter:

$$d_c = 2\sqrt{\frac{A_c}{\pi}}$$

Contour Length: $L_c = \sum_{Cell_c} |x_{i+1} - x_i|$



$$Cell_c = \{x | M(x) = c\}$$



Mask $M(x) = c$
Cell ID c
Pixel position x

Cell Count:

$$N_{cells}$$

Mean Area:

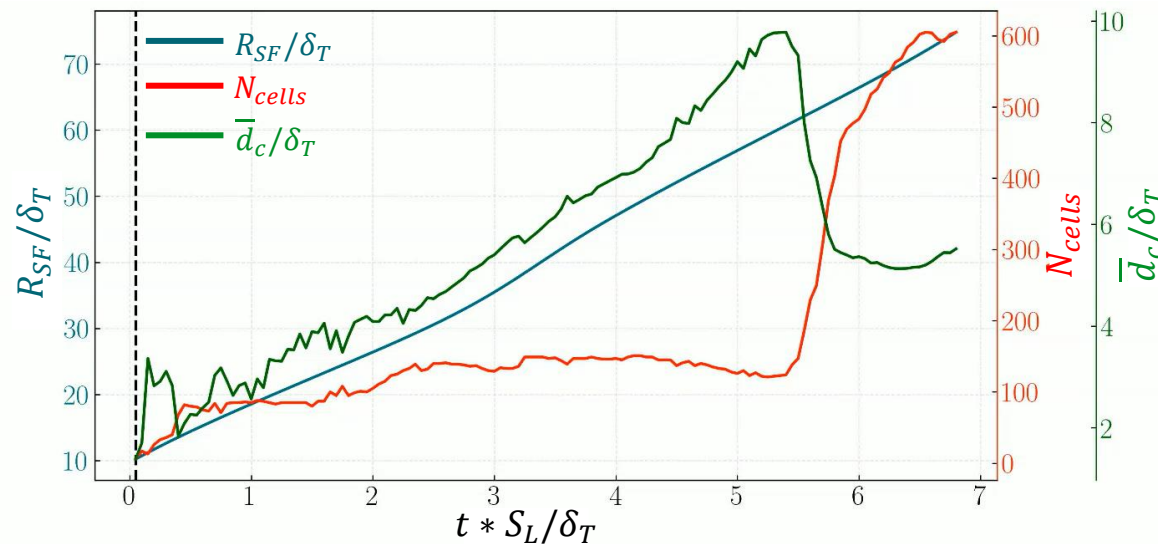
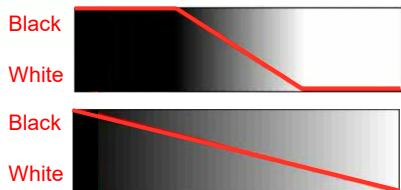
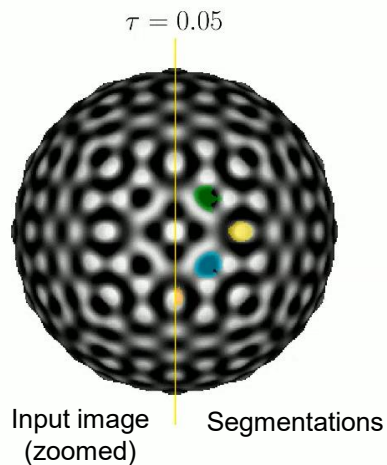
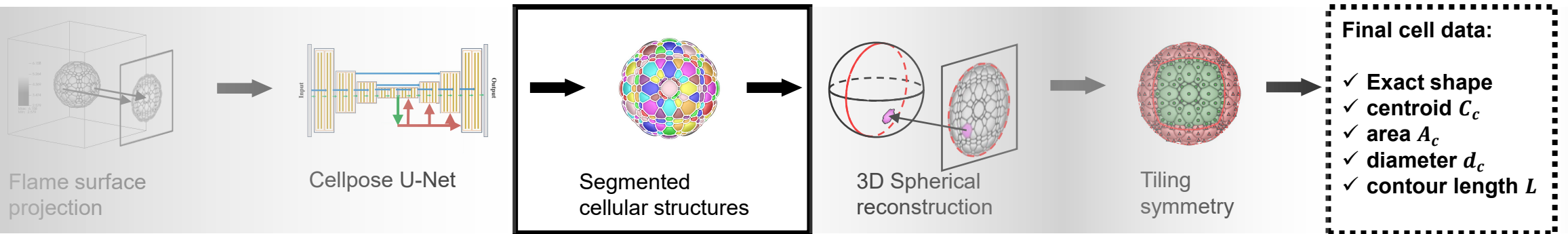
$$\overline{A_c} = \frac{1}{N_c} \sum_c A_c$$

Mean Diameter:

$$\overline{d_c} = \frac{1}{N_c} \sum_c d_c$$

Total Contour Length: $L_c = \sum_c L_c$

Integrated Analysis Pipeline | Extracted Properties



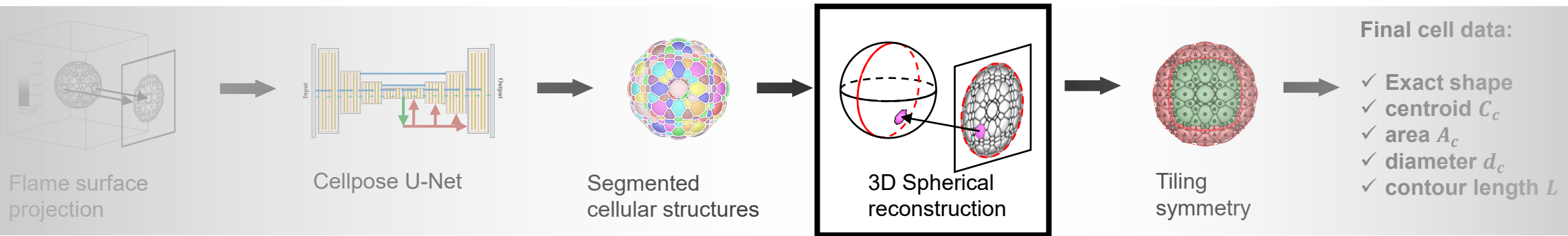
Segmentation issues from:

- Overlapping
- Elongated cells
- Much bigger/smaller than mean cells
- Small pixel size cells
- Soft colour gradients

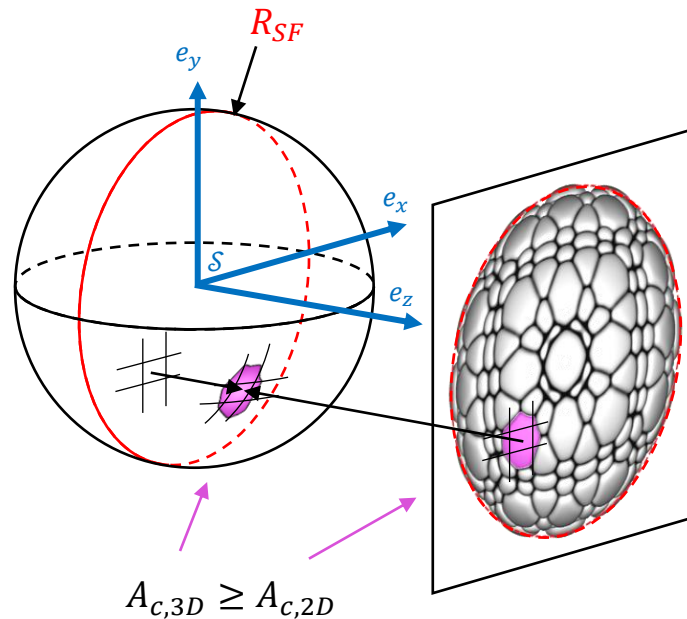
Sensitivity studies found mitigations:

- Sufficient resolution for $d_c \gtrsim d_{trainig} [px]$
- Aggravate colour gradients
- Avoid boundary

Integrated Analysis Pipeline | Spherical Reconstruction



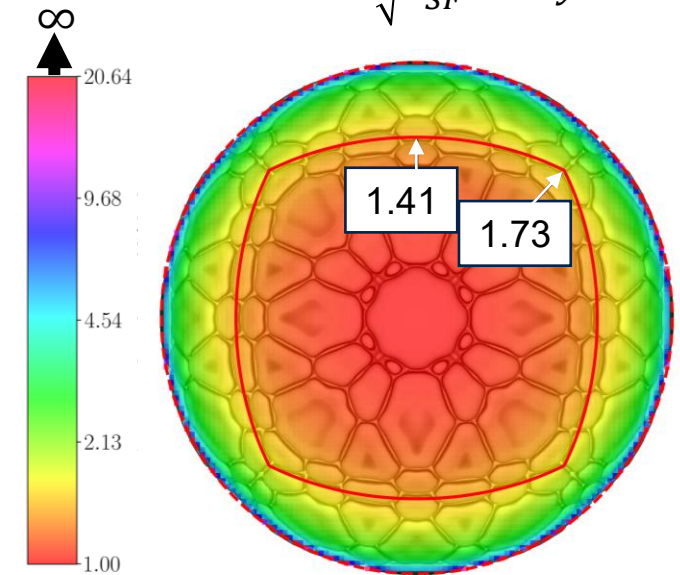
Ideal sphere radius = mean flame surface radius R_{SF}



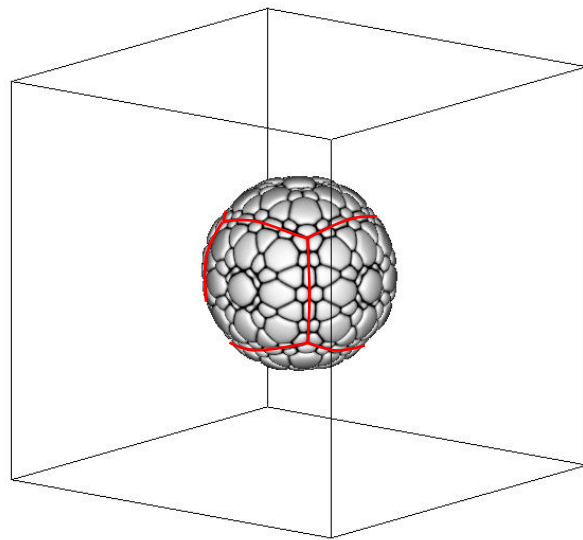
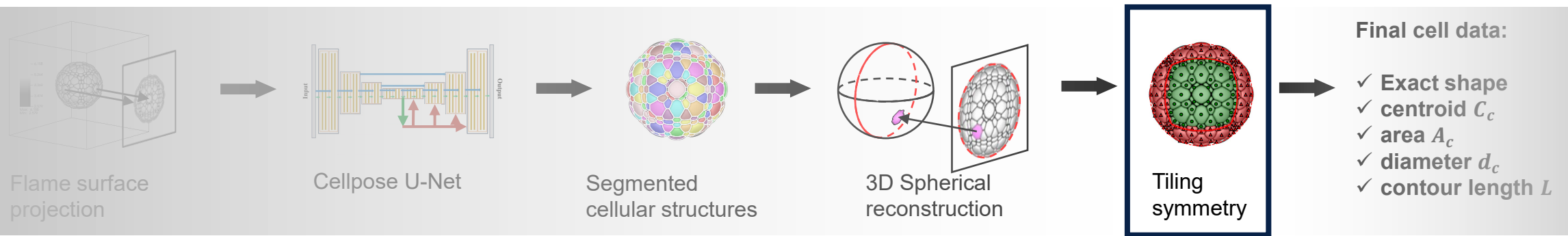
$$A_{c,3D} = \iint_{Cell_c} \frac{R_{SF}}{\sqrt{R_{SF}^2 - x^2 - y^2}} dx dy$$

$$\approx \sum_{Cell_c} \frac{R_{SF}}{\sqrt{R_{SF}^2 - x^2 - y^2}} \times 1px$$

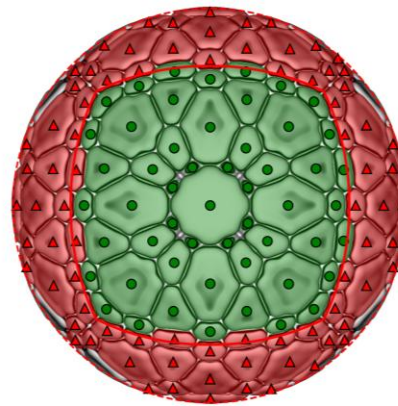
Rescaling Factor: $\frac{R_{SF}}{\sqrt{R_{SF}^2 - x^2 - y^2}}$



Integrated Analysis Pipeline | Tiling from cubed sphere symmetry



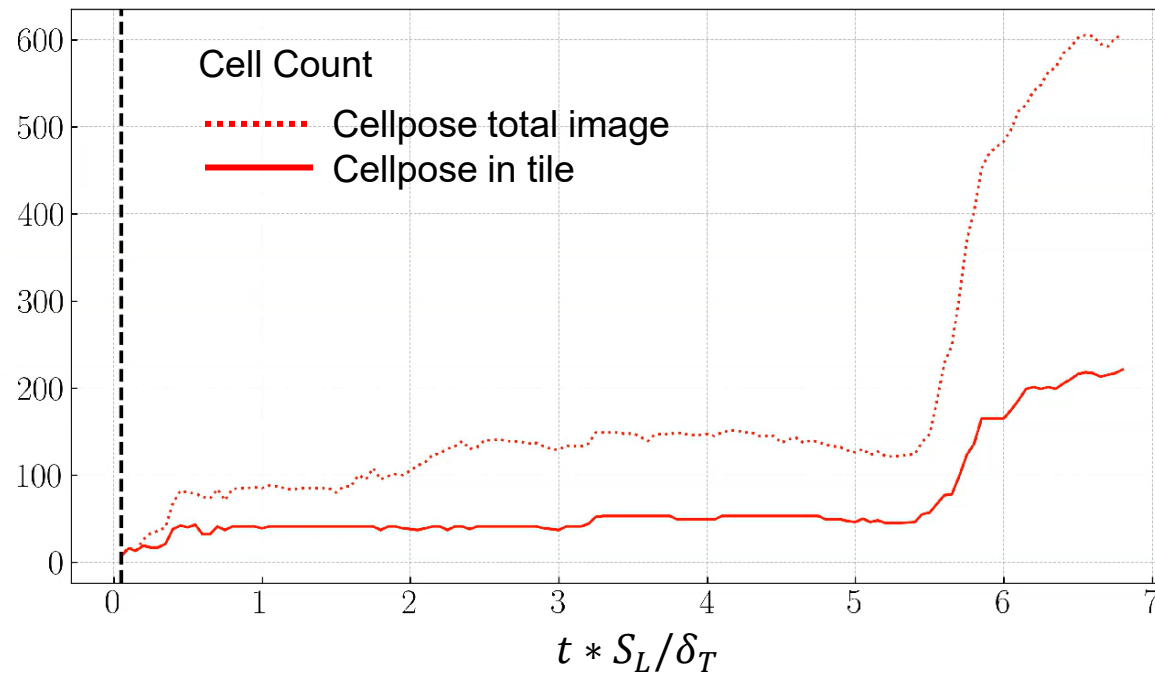
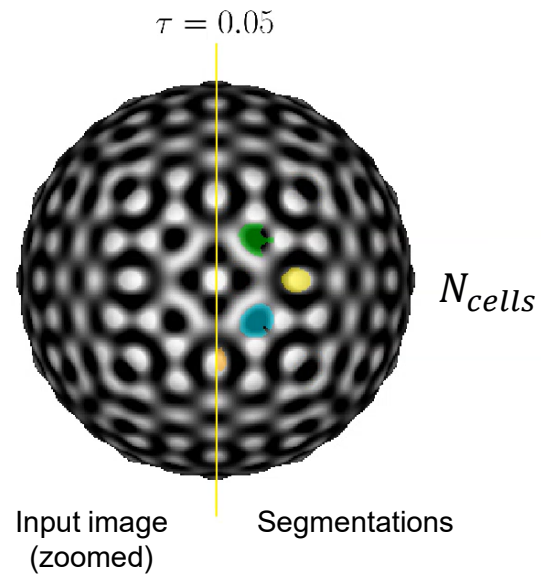
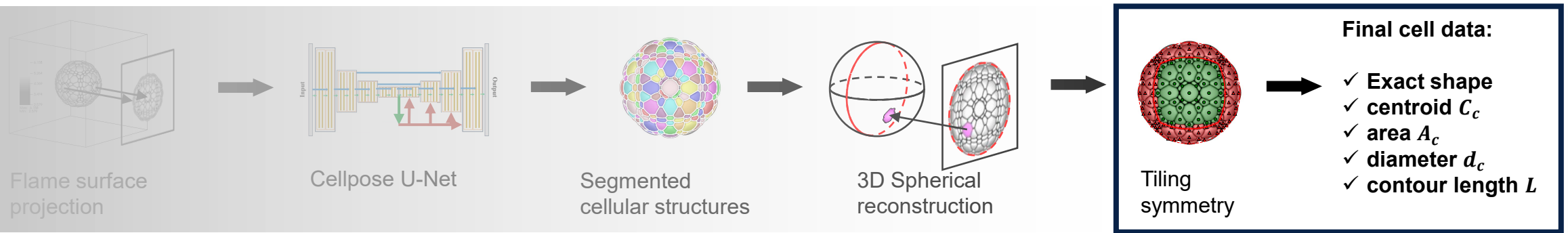
(b)



- Included
- ▲ Excluded

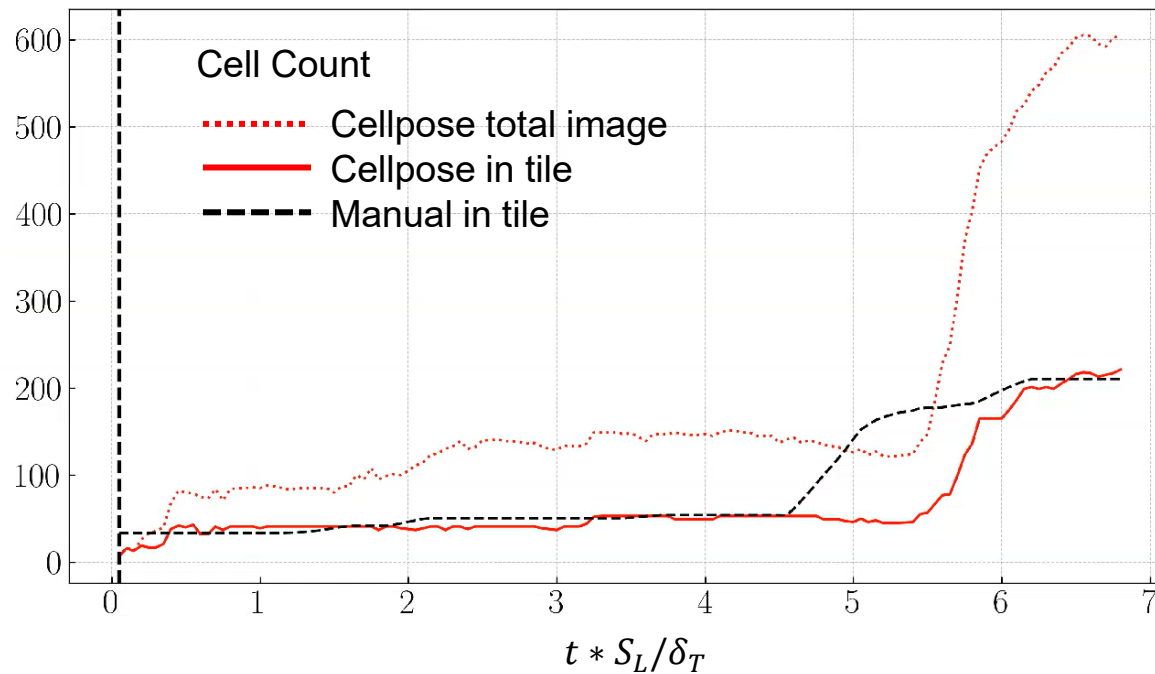
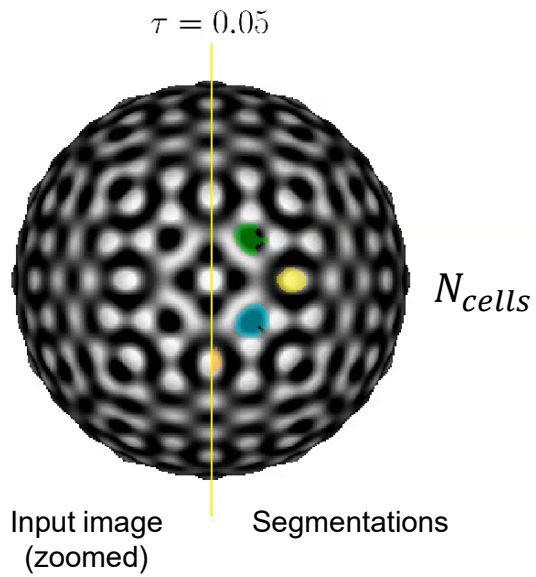
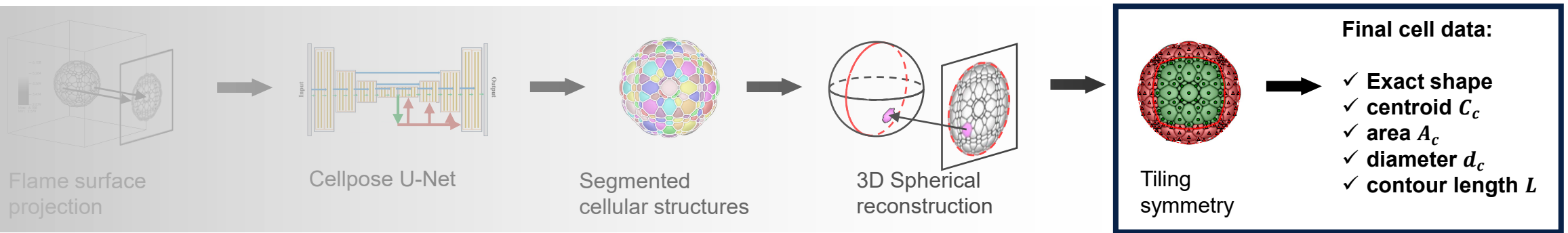
- x6 tiles entire spherical flame
- Avoids boundary prone to segmentation issues
- Avoids diverging 3D rescaling factor

Integrated Analysis Pipeline | Final Cell Count



► Tiling avoids Segmentation issues

Integrated Analysis Pipeline | Final Cell Count

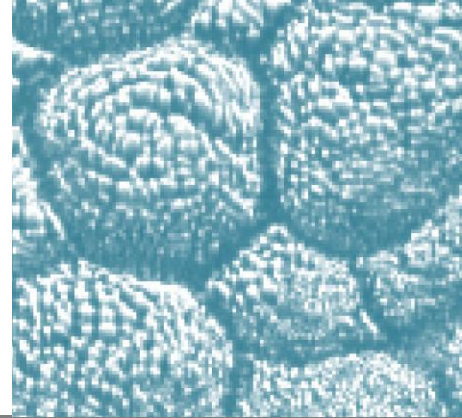


- Tiling avoids Segmentation issues
- Manual count agrees
- Subjective split distinction

What is a cell?

→ Expert trained segmentation model

Results



Results

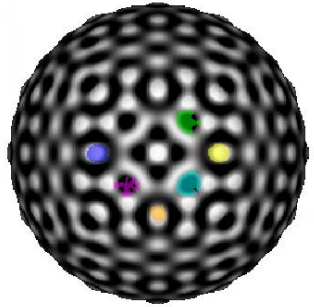
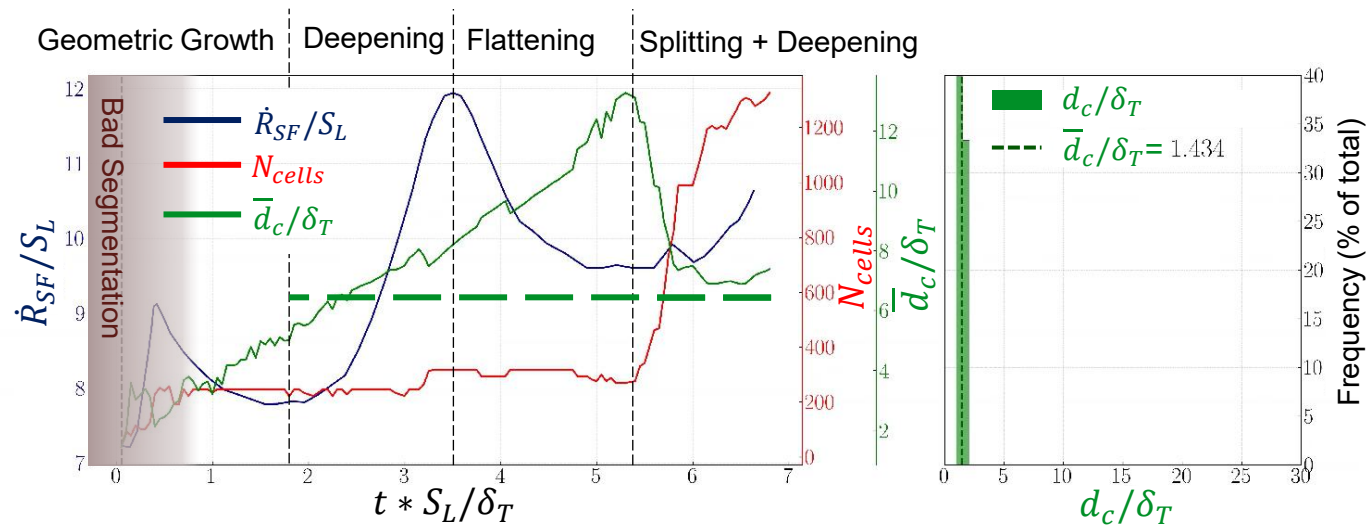


Image + Segmentations



► pre-split $\bar{d}_c \propto t$

► Cell depth information missing

Emerged cell diameter after split

$$\bar{d}_{c,split} \approx 7.2\delta_T$$

Similar to first cell deepening

$$\bar{d}_{c,split} \approx \bar{d}_{c,deepening}$$

Results

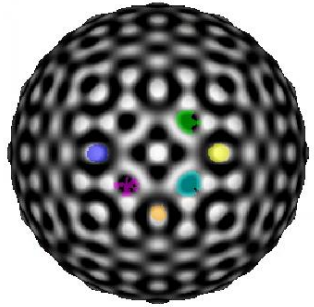
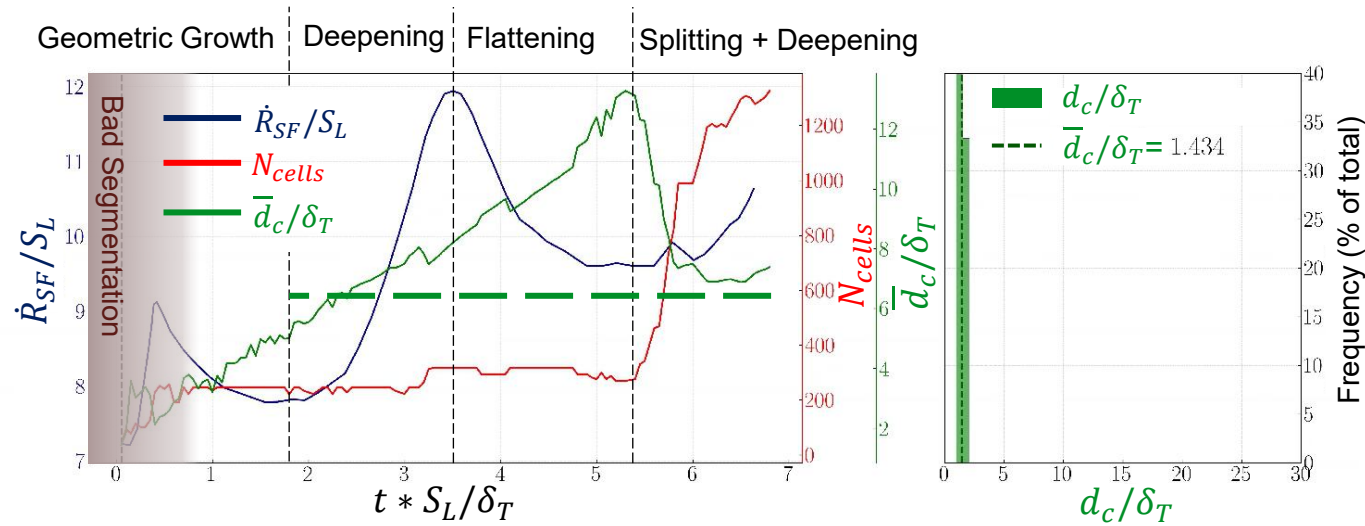


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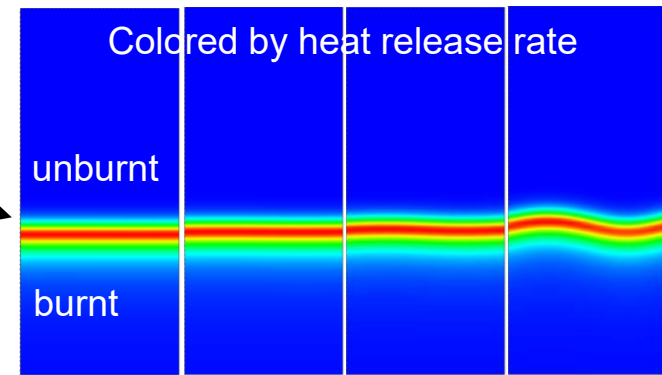
$$\bar{d}_{c,split} \approx \bar{d}_{c,deepening}$$

Fastest growing (2D) wavelength

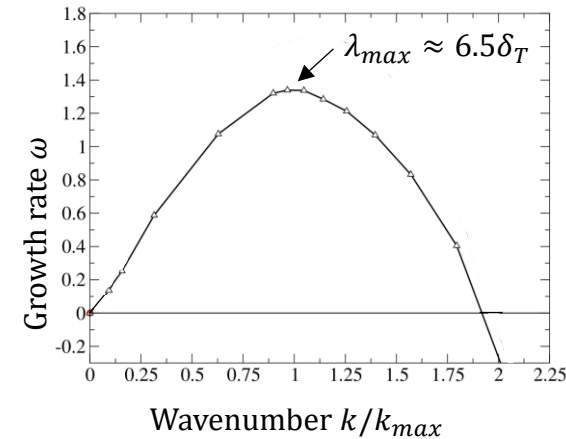
$$\lambda_{max} \approx 6.5\delta_T$$

► $\bar{d}_{c,split} \approx \bar{d}_{c,deepening} \approx \lambda_{max}$

2D channel laminar premixed flame DNS



Altantzis (2011)



small initial perturbation

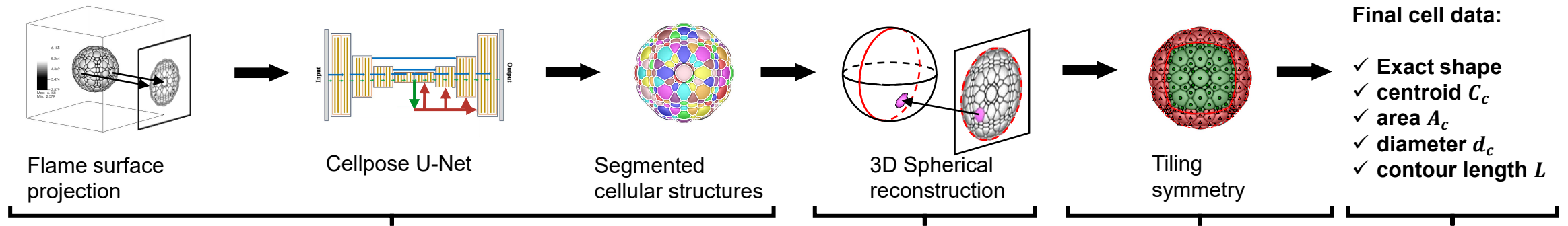
$$x_0 = A_0 \sin ky$$

initially growing as

$$A(t) \sim e^{\omega t}$$

Conclusion

Conclusion



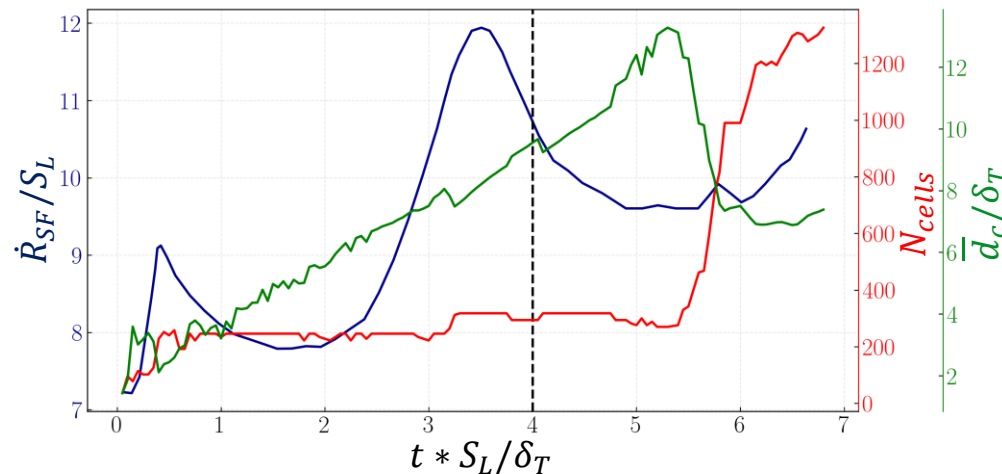
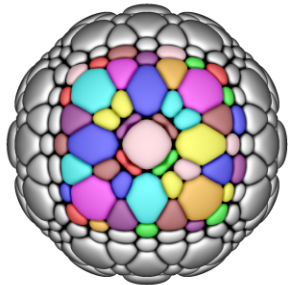
Works well when respecting:

- Resolution: $d_{cell} \gtrsim d_{training} [px]$
- Colour table: steep gradients
- Issues: overlap, elongated cells, large size deviations

► Undo projection errors

► Reduces errors
► Tiling full surface

► One-click results



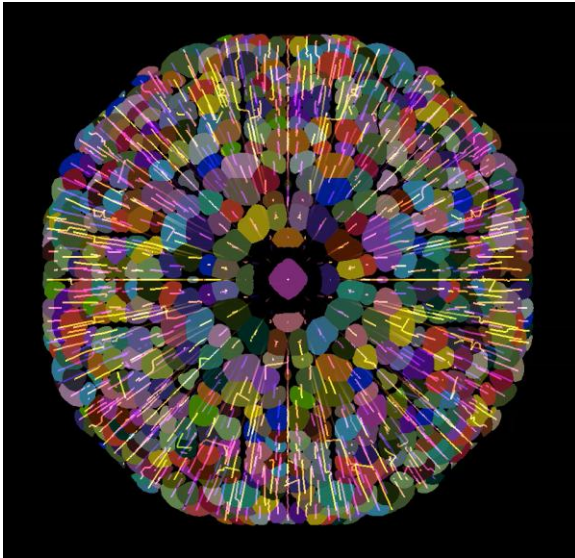
► $\bar{d}_c \propto t$

► Method missed cell depth

► $\bar{d}_{c,split} \approx \bar{d}_{c,deepening} \approx \lambda_{max}$

Further Work

Cell Tracking

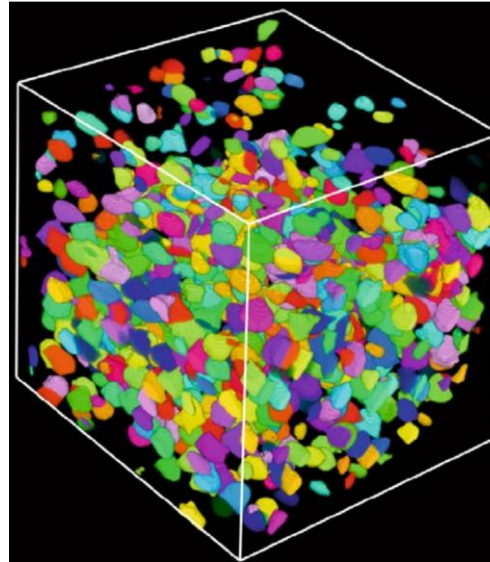


Using "Trackastra" by Galluser et al. (2024)

- Custom Training
- Try the newly released Cellpose SAM
 - More stable on various cell sizes
 - More generalist on various cell types

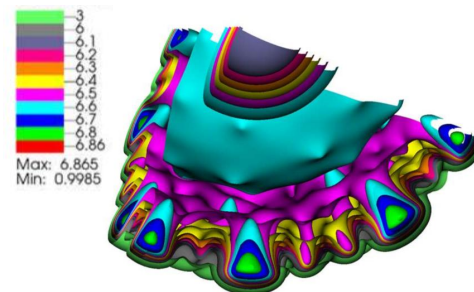
Methods

3D Segmentation

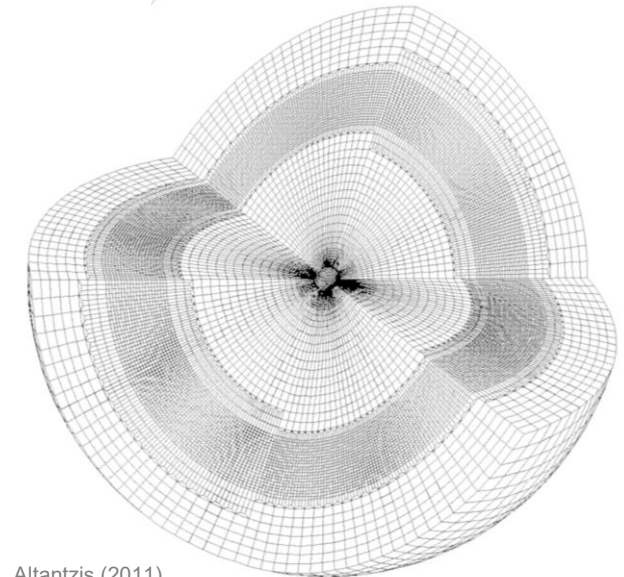


Stringer et al. (2021)

T Isosurfaces



Extend simulation domain – multiple cell splits



Altantzis (2011)

- Deepen physical interpretation
- Different flame surface definitions and coloring
- Different cases: inwards burning spherical flame

Physics

References

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